

Introduction / Problem

Despite the substantial progress in video understanding, modern vision systems still struggle with maintaining object identity across time. Most object detection models process videos as a sequence of independent frames and focus primarily on what is visible in each frame. When an object becomes temporarily occluded, such as when it moves behind another object or leaves the camera's field of view, these systems often fail to maintain a stable presentation of that object. As a result, the object may appear to disappear, lose its identity, or be introduced as a new object when it reappears.

This limitation reflects the challenge of object permanence, a concept from cognitive science describing the ability to reason that an object continues to exist even when it is not directly observable (Baillargeon, 1987). Human naturally maintain object permanence in dynamic environments, but many computer systems lack mechanisms for preserving object identity once visual evidence is temporarily missing.

Standard object detectors and trackers typically operate at the frame level and rely heavily on immediate visual evidence. When detections are absent due to occlusion or detection failure, these systems often terminate the object track or assign a new identity once the object reappears. This behavior results in fragmented trajectories and inconsistent temporal reasoning across frames (Bewley et al., 2016).

These failures highlight that object permanence is not primarily a detection problem but rather a temporal reasoning problem. Improving per-frame detection accuracy alone does not address the underlying issue. Therefore, systems must integrate past observations and maintain memory of objects across time. Without such mechanisms, even highly accurate detectors struggle to maintain coherent object identity in dynamic scenes.

Objective

The objective of this project is to improve object permanence in video understanding by introducing a temporal identity layer that operates on top of an existing object detection system. Rather than retraining the detector, our approach focuses on preserving object identity across frames through post-detection reasoning.

Specifically, we develop a two-stage pipeline that first extracts multi-layer visual embeddings from YOLOv8 detections and then performs temporal linking across frames using similarity-based matching. To recover fragmented tracks caused by occlusion, the system performs an additional relinking stage that uses DINO-based appearance embeddings to verify identity across gaps.

Through this approach, we aim to evaluate whether a lightweight identity reasoning layer can maintain object identity across occlusion events and improve tracking consistency without modifying the underlying detection model.

an explicit post-detection reasoning layer that maintains object memory across occlusion, without retraining existing vision models. We aim to demonstrate that logic-based temporal reasoning applied after object detection can reduce false object disappearances and improve consistency of object identity across video frames.

Background & Related Work

Slot Attention

Slot Attention is an object-centric representation framework that learns to decompose visual scenes into a fixed set of latent “slots,” each intended to correspond to a distinct object (Locatello et al., 2020). In video settings, Slot Attention-based models can be extended to encourage temporal consistency, allowing slots to loosely track objects across frames. This line of work demonstrates that representing scenes as collections of objects rather than monolithic feature maps improves compositionality and interpretability. However, object permanence in these models is implicit: slot identity is not guaranteed across occlusion, and objects may disappear from slots when visual evidence is weak. Slot Attention relies on end-to-end training on large datasets and does not explicitly distinguish between an object being occluded versus no longer existing, limiting its suitability for modular, post-detection reasoning.

Loci and Loci-Looped

Loci introduces an object-centric video model that explicitly separates object identity (“what”) from object location (“where”), enabling more stable object tracking over time (Traub et al., 2023). Loci-Looped extends this framework by incorporating an internal imagination loop that predicts object states during occlusion and gates sensory input against these predictions, allowing the model to maintain object representations even when objects are temporarily invisible (Traub et al., 2024). This approach provides strong evidence that object permanence emerges when perception is explicitly balanced with

prediction and memory. Despite its conceptual relevance, Loci-Looped requires self-supervised training on video data, relies on learned latent slots rather than explicit detections, and involves complex architectures that are difficult to integrate into existing object detection pipelines without retraining.

Graph-Structured World Models (G-SWM)

Graph-Structured World Models (G-SWM) represent scenes as graphs in which nodes correspond to objects and edges encode interactions, learning object dynamics and relational structure from video in a self-supervised manner (Kipf et al., 2020). By modeling objects as entities with state and predicting their evolution over time, G-SWM highlights the importance of structured representations for reasoning about physical dynamics. However, object identity and existence are learned implicitly, and occlusion is handled indirectly through prediction loss rather than explicit reasoning. Like other learned world models, G-SWM requires substantial training and does not provide a clear mechanism for distinguishing between occlusion and disappearance, making it less suitable for deployment as a lightweight, post-detection reasoning layer.

Relationship with our project

This project draws conceptual guidance from several object-centric approaches to object permanence while adopting a modular implementation built on top of an existing object detector. Slot Attention (Locatello et al., 2020) motivates the importance of representing scenes as collections of discrete objects rather than holistic frame representations. This perspective informed our decision to reason explicitly at the object level by treating each detection as an entity with a persistent identity across frames.

The Loci framework (Traub et al., 2023) highlights the value of separating object identity from spatial location to improve temporal consistency in video understanding. This idea aligns with our approach of maintaining object identity through visual embeddings while simultaneously enforcing spatial plausibility during frame-to-frame matching. Loci-Looped further shows that object permanence can emerge from balancing perceptual input with predicted object states during occlusion (Traub et al., 2024). While our system does not implement a learned imagination loop, it similarly attempts to preserve identity through periods of missing detection by reconnecting fragmented tracks during a relinking stage.

Finally, Graph-Structured World Models (Kipf et al., 2020) emphasize the value of structured object representations and temporal reasoning in predictive environments. Inspired by this perspective, our system maintains structured track fragments and

evaluates identity consistency across time using similarity-based linking. Unlike these learning-based approaches, however, our method focuses on lightweight post-detection pipeline that extracts embeddings from YOLOv8 detections and links them across frames using similarity matching and relinking strategies.

System Overview

Our system introduces a modular identity-preserving tracking layer that operates on top of a standard object detector. Instead of modifying the underlying detection model, the system extracts visual representations from detections and links them across time to maintain object identities. The overall pipeline is designed to preserve object identity through occlusion events and to reconnect fragmented tracks after temporary detection failures.

The system operates as a two-stage offline pipeline. The first stage enriches raw detections with feature representations while the second stage performs temporal linking to construct coherent object tracks across frames.

Stage 1: Trace Enrichment

In the first stage, the system processes each frame of the input video using the YOLOv8 object detector. For every detected object, the detector produces bounding boxes, class predictions, and confidence scores. In addition to those outputs, our system extracts intermediate activation features from multiple layer of the YOLO network.

These features capture different levels of visual information. Early layers encode appearance cues such as color and texture, intermediate layers capture structural and spatial information about objects, and later layers contain class-level semantic signals. To combine these complementary signals, we construct a composite embedding from selected YOLO layers and project the resulting feature vector into a lower-dimensional space using Principal Component Analysis (PCA). The resulting embedding serves as a compact identity representation for each detected object.

Alongside the YOLO embeddings, the system also extracts appearance descriptors from a DINO vision transformer. These embeddings are stored as auxiliary metadata and are used later during the relinking stage to verify identity across occlusion gaps.

Stage 2: Temporal Linking

After detections are enriched with embeddings, the second stage links detections across frames to construct object trajectories. For each pair of consecutive frames, the system compares the embeddings of detected objects using cosine similarity. Candidate matches are filtered using spatial plausibility constraints to ensure that objects are only linked if their movement between frames is physically reasonable.

The final assignment between detections and existing tracks is solved using the Hungarian algorithm, which enforces a globally optimal one-to-one correspondence between objects in consecutive frames. This process builds object tracks over time and allows the system to maintain consistent identities as objects move through the scene.

To manage track lifecycle, each object track transitions through a small state machine. Newly detected objects begin as tentative tracks, become active once they accumulate sufficient evidence, transition to a lost state when detections are temporarily missing, and are eventually closed when they can no longer be extended.

Relinking Across Occlusions

Occlusion events can cause tracks to fragment when an object temporarily disappears from detector's output. To address this, the system performs a secondary relinking pass after the initial temporal linking stage. During this step, closed track fragments are compared to determine whether they correspond to the same underlying object.

Identity verification during relinking uses DINO embeddings, which provide stronger appearance discrimination than YOLO features alone. For each track fragment, the system builds a gallery of representative appearance embeddings collected across its lifetime. Fragment pairs are then compared by computing similarity between their galleries, allowing the system to reconnect tracks that were split due to occlusion.

If appearance evidence is insufficient, the system falls back to similarity computed from YOLO embeddings or spatial plausibility checks. Accepted fragment pairs are merged into a single canonical track, producing the final identity-consistent trajectories.

Together, these stages create a lightweight post-detection tracking system that preserves object identity across frames while remaining compatible with standard object detection pipelines.

Methodology

Our system is designed as a post-detection identity-preserving tracking pipeline that operates on top of YOLOv8. Rather than training a new tracker end-to-end, we treat detection as the perceptual front end and build a separate identity reasoning layer that links detections across time. The overall methodology consists of five main components: multi-layer feature extraction from YOLOv8, dimensionality reduction with PCA, frame-to-frame matching using cosine similarity and spatial plausibility constraints, track management through a state machine, and a relinking stage using DINO embeddings to recover tracks fragmented by occlusion.

Multi-Layer YOLO Feature Extraction

The first step in the pipeline is to build an identity representation for each detection. A standard detector output, such as bounding box coordinates, class label, and confidence score, is not sufficient for robust identity preservation because those values are primarily designed for localization and classification rather than re-identification. To address this, we extract intermediate activations from several layers of YOLOv8 for every detected object.

Layer	Role
4.cv1	Appearance features
15	Semantic structure
22.cv3.0	Class-level signals

These layers were chosen to capture complementary information. Early backbone layers preserve low-level visual cues such as texture, edges, and color patterns. Mid-level layers provide more structured object information, such as shape and part configuration, which tends to remain more stable across viewpoint changes. Detection-head layers encode the model’s class-discriminative representation, which can help maintain category consistency, although this signal is less useful for distinguishing between multiple instances of the same class.

Instead of relying on a single layer, we combine all three into a composite embedding. This is important because identity preservation across occlusion depends on multiple kinds of evidence. Appearance cues may help distinguish visually different objects, semantic cues may remain stable when exact pixel values change, and class-level cues can act as a weak regularizer so that identity assignment does not drift arbitrarily.

Weight Selection for the Composite Embedding

The three YOLO feature sources are not equally informative, so the system assigns them different weights. The final weights used in the project were:

Layer	Role	Weight
4.cv1	Appearance features	0.549
15	Semantic structure	0.351
22.cv3.0	Class-level signals	0.100

These weights were selected through a separability sweep over candidate YOLO layers. The purpose of the sweep was to measure how well each layer's activations separated detections that should be considered different from those that should be considered similar. In other words, the sweep estimated how useful a given layer was as an identity descriptor.

- The appearance layer received the highest weight because early visual features were the strongest signal for distinguishing objects based on texture and local appearance.
- The semantic layer received the second highest weight because it captured more stable structural information that remained useful when appearance changed slightly across frames.
- The class-level layer received the smallest weight because class logits are helpful for category consistency but often do not distinguish two visually similar instances of the same class well.

A key caveat is that the calibration sweep was not fully instance-supervised. The sweep used class fallback rather than stable instance IDs, meaning the separability scores reflect class separation more strongly than true same-instance re-identification. As a result, these weights should be interpreted as a practical empirical choice for this system rather than a definitive instance-optimal solution. Even with this limitation, the weighted multi-layer representation provided a better identity signal than relying on a single YOLO layer alone.

Dimensionality Reduction with PCA

After feature extraction, the raw composite embedding is still relatively high-dimensional. The three selected layers together produce a 208-dimensional YOLO feature vector before projection. Working directly in this space is possible, but it has several drawbacks: increased computational cost, amplified noise, and less stable similarity comparisons.

To make the representation more compact and better conditioned for matching, we apply Principal Component Analysis (PCA). PCA is a linear dimensionality reduction method that finds orthogonal directions of maximum variance in the data and projects the original features onto those directions. In effect, it keeps the most informative variation while discarding redundant or low-variance dimensions.

In our pipeline, PCA serves two purposes. First, it compresses the embedding into a lower-dimensional identity space with a target dimension of 128. Second, it improves the quality of similarity comparisons by concentrating the most informative structure into fewer dimensions. The effective output dimension may be lower than 128 in smaller runs, since PCA cannot produce more meaningful components than the sample count allows. This projected embedding is then normalized and used as the primary identity descriptor for frame-to-frame matching.

Frame-to-Frame Matching with Cosine Similarity

Once every detection has an embedding, the next task is to determine whether a detection in the current frame corresponds to an existing track from previous frame. We treat this as a similarity-based assignment problem.

The primary matching signal is cosine similarity between normalized embeddings. Cosine similarity measures the angle between two vectors rather than their raw Euclidean distance. This makes it well suited to identity matching because we care more about whether two detections point in a similar representational direction than whether their raw magnitudes match exactly. In practice, cosine similarity is commonly used for retrieval and re-identification because it is stable under feature scaling and works naturally with normalized embeddings.

A candidate match is considered visually plausible only if its cosine similarity exceeds a predefined threshold. In the final configuration, the similarity threshold was set to 0.70. This value was chosen empirically during system calibration to balance two competing objectives: preventing identity switches between visually similar objects while still allowing legitimate matches when object appearance varies slightly between frames.

Lower thresholds tended to admit more incorrect matches, particularly in scenes containing multiple objects with similar appearance or in cases where detector noise produced visually ambiguous detections. Conversely, higher thresholds often caused legitimate matches to be rejected, which increased track fragmentation when objects experienced small appearance changes or partial occlusion. Through experimentation across several video scenarios, a threshold of 0.70 provided the most stable trade-off,

preserving consistent object identities while minimizing incorrect cross-object assignments.

Spatial Plausibility Constraint

Visual similarity alone is not enough, especially in scenes where multiple objects share similar appearance or where detector produces spurious detections. To reduce incorrect identity swaps, the system also applies spatial plausibility constraint before assigning matches.

The main spatial rule uses centroid distance. For a track in the previous frame and a candidate detection in the current frame, the Euclidean distance between their bounding box centers is computed and normalized by the frame diagonal. A match is only considered if this normalized centroid distance is less than or equal to a threshold. In the project configuration, this threshold was 0.40 of the frame diagonal.

This gate encodes a simple physical prior: an object should not teleport across the scene from one sampled frame to the next. By combining appearance similarity with spatial plausibility, the system avoids many impossible or highly unlikely matches. This is especially important because the tracker is intended to preserve identity, not merely to connect visually similar detections.

One-to-One Assignment with the Hungarian Algorithm

After filtering candidate matches by similarity and spatial plausibility, the remaining problem is to choose the best assignment between detections and existing tracks. This cannot be solved greedily because assigning one detection to one track may block a better overall set of matches.

To enforce globally consistent matching, the system uses the Hungarian algorithm. The Hungarian algorithm solves the linear assignment problem: given a cost matrix between tracks and detections, it finds the one-to-one matching that minimizes total cost or equivalently maximizes total similarity after conversion. In this project, the matching matrix is derived from embedding similarity, with penalties applied when necessary.

The Hungarian algorithm is appropriate here for two reasons. First, it guarantees that each detection is linked to at most one track and each track is extended by at most one detection per frame. Second, it finds a globally optimal assignment among all feasible pairings instead of making local greedy decisions that may cause track fragmentation or identity switches later.

This step is the core of the frame-to-frame linker. Once the assignment is computed, matched detections extend existing tracks, and unmatched detections or tracks are handled by the track state machine.

Class Mismatch Handling

Although YOLO outputs a predicted class for each detection, the system does not treat class labels as a hard identity rule. This is because detector labels can be noisy, especially under occlusion or low confidence. Instead, class disagreement is handled using a soft penalty rather than absolute rejection.

During frame-to-frame matching, cross-class candidates may still be considered, but they receive a configurable class mismatch penalty. A similar but smaller penalty is also used in relinking. This design choice reflects the fact that a track can remain identity-consistent even when the detector temporarily mislabels it. Over the full lifetime of a track, the final displayed class is resolved by a confidence-weighted vote across all linked detections rather than by trusting a single frame's class prediction.

Track State Machine

To manage object identity over time, each track is governed by a four-state lifecycle:

TENTATIVE -> ACTIVE -> LOST -> CLOSED

Each state has a specific function in controlling how much evidence is required before trusting a track and how long the system should preserve it after detections disappear.

- **TENTATIVE:** A new detection begins as a tentative track. At this point, the system has not yet accumulated enough evidence to conclude that it represents a stable object trajectory. This prevents transient false positives from immediately becoming full tracks.
- **ACTIVE:** Once a tentative track receives enough consistent support, it becomes active. Active tracks are considered valid identities and participate in normal frame-to-frame assignment.
- **LOST:** If an active track is temporarily unmatched, it transitions to lost rather than being immediately deleted. This state is critical for handling brief occlusions, detector dropouts, or sampling gaps. The track still exists in memory and may be recovered in subsequent frames.
- **CLOSED:** When a lost track is no longer eligible for direct extension, it becomes closed. Closed tracks are not extended during ordinary frame-to-frame linking, but they remain available as candidates during the relinking phase.

This state machine provides a practical approximation of object persistence. It prevents immediate deletion when evidence disappears while still imposing enough structure to avoid keeping every object alive forever. The system also maintains multiple descriptors for each track, including recent embeddings, exponential moving average representations, and historical vectors so that identity matching remains stable even when appearance shifts over time.

DINO Sidecar Embeddings for Reasoning

Frame-to-frame matching works well when occlusions are short and detections remain relatively continuous. However, longer occlusions can fragment a single object trajectory into multiple closed track segments. To repair these splits, the system performs a second-stage relinking pass.

For relinking, the pipeline uses embeddings from DINO, a self-supervised vision transformer. DINO embeddings are not used during the main per-frame linking stage. Instead, they are stored as sidecar metadata during enrichment and reserved for relinking, where discriminative appearance comparison matters more than local frame-to-frame stability. This design keeps the main linker lightweight while using a stronger appearance model only when needed.

The motivation is that YOLO features are optimized for detection, not necessarily for instance re-identification across large temporal gaps. DINO embeddings provide a more robust appearance-based signal for deciding whether two closed fragments belong to the same underlying object after an occlusion.

Gallery Construction and Top-k Matching

Each closed fragment accumulates valid DINO observations over its lifetime. At relink time, these observations are reduced into a fixed-size representative gallery. The gallery is built using temporal bucketing: the fragment lifetime is divided into temporal segments, and the highest-confidence DINO embedding from each segment is retained.

This design is important because it preserves temporal diversity. If the gallery contained only the highest-confidence frames overall, it might overrepresent one short part of the fragment and fail to capture appearance variation across time. By spreading samples across the fragment lifetime, the gallery better represents the object's full observed appearance.

To compare two fragments, the system computes the pairwise cosine similarity matrix between their DINO galleries. Instead of reducing this matrix to a single mean over all pairwise comparisons, it takes the top-k strongest matches and averages them. This top-k

mean reduces the influence of noisy or unrepresentative frames and emphasizes whether the fragments agree on several strong appearance correspondences rather than on just one accidental match. The representative gallery size defaults to 20 and the top-k reducer defaults to 3. If the relink score exceeds the DINO relink threshold, the two fragments are merged into a single canonical track.

Fallback Relinking

Not all fragments have sufficient DINO coverage, and not all relink decisions should depend on one feature type alone. For this reason, the system uses a fallback hierarchy:

1. DINO-based relink when both fragments have enough valid DINO samples and DINO relinking is enabled.
2. YOLO-based relink when DINO evidence is unavailable or insufficient.
3. Spatial fallback for unresolved fragment pairs.

This layered strategy improves robustness. DINO serves as the strongest identity signal during occlusion recovery, but the system can still attempt repair when DINO is missing by using the existing YOLO embedding representation or motion/spatial plausibility cues. Accepted relink edges merge fragmented tracks into a single final identity path.

Experimental Setup

All experiments were conducted on two distinct video domains: controlled ball-tracking scenarios and real-world motion sequences drawn from the DAVIS benchmark. The ball-tracking scenarios consist of eight short clips designed to isolate specific tracking challenges, including lateral motion, occlusion with ball removal, and bounce-back trajectories. The DAVIS stress subset was constructed using a curated selection of nine sequences chosen to stress the tracker across diverse object classes, motion patterns, and occlusion conditions.

All experiments use the R4 configuration throughout. The composite YOLO embedding is drawn from layers 4.cv1, 15, and 22.cv3.0 with weights 0.549, 0.351, and 0.100 respectively, producing a 208-dimensional raw feature vector projected to a 128-dimensional target space via PCA. DINO CLS vectors of dimension 384 are extracted as sidecar metadata and used exclusively during relinking. The activation top-k is set to 64. Frame-to-frame matching uses a cosine similarity threshold of 0.70 and a maximum normalized centroid distance of 0.40. Relinking is performed with DINO enabled at a threshold of 0.55, a YOLO relink threshold of 0.55, a spatial fallback threshold of 0.40, and

an unlimited relink gap (`relink_max_gap_frames = -1`). The DINO gallery size is 20 with a top-k reducer of 3.

For the DAVIS subset, videos are processed at a dense sample rate of 1 (every frame). Track and class evaluation follow fixed heuristics: a sequence passes class evaluation when at least 85% of detections fall within the expected class set, and passes track evaluation when the strongest expected-class track covers at least 50% of sampled frames. A sequence is marked as overall passing only when both criteria are satisfied.

Results

Ball Tracking

Across the eight ball-tracking scenarios, the pipeline processed 427 frames and produced 657 detections. Each scenario contained a single ground-truth ball object. The system produced a total of 55 tracks across all scenarios, of which 38 were valid, with 9 relink edges accepted across the full suite.

The DINO relink threshold sweep (R0–R6) reveals that enabling DINO relinking at thresholds between 0.40 and 0.55 (R1–R4) consistently reduces total track count from 56 to 55 and maintains 9 relink edges, compared to 8 in the DINO-disabled baseline (R0). At thresholds of 0.60 and above (R5–R6), DINO coverage remains high but accepted relink edges fall back to 8, matching the baseline. R4 (`relink_dino_threshold = 0.55`) is selected as the operating configuration as it achieves the maximum relink benefit while applying the most conservative threshold within the winning range.

Qualitatively, the system successfully maintains a single canonical ball identity across the full 10sec_Left_to_Right clip and correctly recovers the ball trajectory after a direction reversal in Left_bounce_back. Failure cases are detector-driven: in Occlusion_ball_removed and No_occlusion_ball_removed, persistent YOLO false positives form coherent tracks that survive relinking despite carrying incorrect semantics.

DAVIS Stress Benchmark

On the nine-sequence DAVIS stress subset, the pipeline achieves an overall pass on six sequences: parkour, bmx-bumps, breakdance, dance-twirl, bear, and motocross-jump. Three sequences fail: horsejump-high, drift-chicane, and scooter-black.

Passing sequences demonstrate that the system maintains strong track continuity for dominant objects across occlusion and viewpoint change when the detector produces consistent class labels. In parkour, the primary person track spans 96 of 100 sampled frames. In motocross-jump and dance-twirl, expected-class detection purity exceeds 95%.

Failing sequences expose the boundary between linker capability and detector reliability. In drift-chicane, the primary race car track covers 47 of 52 frames, indicating strong spatial continuity, but overall expected-class share falls to 32.3% due to widespread label drift into person, truck, boat, and airplane categories. In scooter-black, a single physical rider-and-scooter entity is fragmented into parallel motorcycle, person, and car tracks, representing a case where temporal consistency alone cannot resolve conflicting class identity assigned by the detector. In horsejump-high, scene clutter produces spurious detections labeled as potted plant, stop sign, and car, depressing expected-class share below the 85% threshold despite a fully stable rider track.

Discussion

The results across both evaluation domains support the central claim of this project: a lightweight post-detection identity reasoning layer can meaningfully improve object permanence in video without modifying the underlying detector. In the ball-tracking domain, enabling DINO-based relinking recovers additional track merges that the YOLO-only baseline misses, reducing fragmentation while preserving specificity at the chosen threshold. In the DAVIS domain, the system maintains stable primary-object tracks in the majority of sequences, including cases involving heavy motion blur, crowd occlusion, and scale change.

At the same time, the results consistently reveal that the primary failure mode is detector-driven rather than linker-driven. When YOLO produces stable, semantically correct detections, the linker constructs coherent long-range trajectories. When YOLO emits persistent false positives or assigns inconsistent class labels, the linker faithfully preserves those errors across time. This asymmetry is visible in drift-chicane, where the strongest car track is spatially correct but the sequence-level class distribution is dominated by off-class labels, and in scooter-black, where the linker cannot unify what the detector treats as separate object categories.

This pattern highlights a structural limitation of any post-detection approach: the identity layer can only reason about what the detector provides. It cannot correct semantic errors

introduced upstream. The class mismatch penalty and confidence-weighted class voting partially mitigate this, allowing tracks to survive brief detector mislabeling, but they do not resolve persistent or systematic label noise.

The DINO threshold sweep also reveals a practically useful result: DINO relinking improves outcomes at moderate thresholds but becomes overly conservative at higher values. The fact that R1 through R4 produce identical aggregate results suggests that the relink decisions are robust within a range, and that the benefit is not sensitive to exact threshold tuning within that window.

Limitations

Several limitations constrain the current system. First, the layer weight calibration is class-level rather than instance-level. The separability sweep used class fallback rather than verified instance identities, meaning the selected weights reliably separate object categories but may not optimally distinguish visually similar instances of the same class. This is a fundamental limitation of the calibration methodology rather than the embedding architecture itself.

Second, the system has no mechanism to suppress or correct persistent false positives from the detector. If YOLO emits a consistent spurious detection across many frames, the tracker will maintain it as a coherent identity. This behavior is internally consistent but semantically incorrect, as demonstrated by the ball-removal scenarios and several DAVIS failure cases.

Third, heavy occlusion remains only partially recoverable. The gallery-based DINO relink stage addresses many fragmentation cases, but cannot resolve situations where an occluder itself generates a competing detection that overlaps with the target object's expected location. In such cases, the tracker may link to the occluder rather than preserving the target identity.

Fourth, the pipeline is offline and assumes that all frames are available for enrichment before linking begins. This design is appropriate for the evaluation setting but precludes direct deployment in real-time applications without architectural modification.

Finally, the evaluation metrics used for the DAVIS subset are practical reporting heuristics rather than standard tracking benchmarks such as MOTA or IDF1. While these heuristics provide interpretable pass/fail summaries, they do not support direct comparison with other tracking systems in the literature.

Conclusion

This project introduced a two-stage offline pipeline for identity-preserving object tracking that operates as a post-detection reasoning layer on top of YOLOv8. By extracting multi-layer YOLO embeddings, projecting them into a compact identity space via PCA, and linking detections across frames using cosine similarity and the Hungarian algorithm, the system constructs temporally coherent object trajectories without modifying the underlying detector. A secondary relinking stage using DINO gallery embeddings recovers tracks fragmented by occlusion, providing a stronger appearance-based identity signal at the points where it is most needed.

Experiments across controlled ball-tracking scenarios and the DAVIS stress benchmark demonstrate that the approach successfully preserves object identity in the majority of evaluated sequences. The primary failure mode is detector-driven: when the detector produces semantically inconsistent labels or persistent false positives, the identity layer propagates rather than corrects those errors. This finding suggests that improvements to detection reliability, particularly class consistency under occlusion and motion blur, would yield the largest downstream gains in tracking quality.

The results also confirm that object permanence in video is not solely a detection problem. Even with a fixed, unmodified detector, the addition of a structured identity reasoning layer meaningfully reduces track fragmentation and improves temporal consistency. This supports the broader thesis that modular post-detection reasoning is a viable and practical path toward more robust video understanding systems.

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